

PA-LOI Framework (Upstream)

Sensors
& HD Map

1. Occlusion Geometry Analyzer

Shadow Regions

2. Dynamic Risk Field Generator

Risk Corridor C_{risk}

3. Risk-Integrated Trajectory
Optimizer (MIND)

Proactive
Safe Trajectory

Reactive Safety

TTC Monitor

$TTC < 1.4s$

Emergency Braking
Actuation

